

Orthogonal Euler Circles as an Observation Frame: The Chinese Remainder Theorem and Prime Observability (Exercise XXV)

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Abstract

We report the third completed exercise of a re-formalization program (“0pat”): the insight that the prime axis can be observed through two orthogonal Euler circles — the periodized number line $\mathbb{Z}/p\mathbb{Z}$ as a circle, two such circles at coprime moduli as an orthogonal coordinate system — is re-stated in pure known mathematics and formally proved in Lean 4 / mathlib with *no* pat framework concepts and *no* complex plane. The re-statement is the algebra of double-modulus observation: two orthogonal circles ($\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z}$) compose into the circle of the product modulus by the Chinese remainder theorem (`ZMod.chineseRemainder`); a prime factor p of a is visible as the zero projection on the $\mathbb{Z}/p\mathbb{Z}$ circle; distinct primes are distinguished by nonzero projection on each other’s circles; a prime observed on a non-own circle is a unit of that circle’s group, satisfying Fermat’s little theorem $p^{q-1} \equiv 1 \pmod{q}$; and on its own circle a prime satisfies the complete Fermat algebra $a^p \equiv a \pmod{p}$. Seven theorems, zero errors, zero sorry, mathlib v4.32.2.

Keywords: Lean, formalization, Chinese remainder theorem, Fermat’s little theorem, circle group, 0pat, reproducibility

1 The observation frame

The starting insight of this exercise is that the prime axis can be observed through circles. The number line, periodized modulo p , is the circle $\mathbb{Z}/p\mathbb{Z}$; the mapping between the real axis and a circle is already formalized in pure known mathematics (stereographic projection $\mathbb{R} \cup \{\infty\} \cong S^1$ and the covering map $\mathbb{R} \rightarrow \mathbb{R}/2\pi\mathbb{Z}$), so observing integers on a circle is a legitimate coordinate choice, not a new structure. The question posed here is whether *two* such circles, orthogonal to each other, form an observation frame for the prime axis.

The answer, in pure known mathematics, is the algebra of double-modulus observation. Two circles at moduli p and q are orthogonal exactly when p and q are coprime: the product circle $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z}$ then composes into the single circle $\mathbb{Z}/pq\mathbb{Z}$ of the product modulus. The composition is the Chinese remainder theorem, present in mathlib as `ZMod.chineseRemainder`: for coprime m, n , the rings $\mathbb{Z}/mn\mathbb{Z}$ and $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ are isomorphic. This is the coordinate law of the frame: a number a has exactly one position in each circle, and the pair of positions determines a modulo the product.

The frame is legitimate because the composition law is an isomorphism: observing through two orthogonal circles loses nothing that observing through the product circle would keep. The frame is useful because prime structure becomes visible in the projections: this is the content of the theorems below.

2 Prime observability in the frame

Seven theorems, all machine-checked, state what the frame sees. Each statement is classical mathematics; the novelty is the exact formulation of the observation and the machine-checked proof.

The coordinate law (`orthogonal_frame_crt`). For coprime m, n , the product of the two orthogonal circles is the circle of the product modulus: `ZMod.chineseRemainder` gives $\mathbb{Z}/(mn)\mathbb{Z} \simeq \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ as rings. This is the law that makes the frame a coordinate system: two orthogonal circle positions determine one position on the product circle, and conversely.

Prime factors are visible (`factor_observable_zero`). If p is prime and $p \mid a$, then a projects to 0 on the $\mathbb{Z}/p\mathbb{Z}$ circle: $(a : \mathbb{Z}/p\mathbb{Z}) = 0$. A prime factor of a is therefore directly observable as a zero in the corresponding coordinate — the first coordinate of the frame detects the factor.

Distinct primes are distinguished (`distinct_primes_observable`). If p and q are distinct primes, then p projects to a nonzero element of the $\mathbb{Z}/q\mathbb{Z}$ circle. The proof is divisibility-theoretic: $q \nmid p$ (a prime divides a prime only if they are equal), hence $p \bmod q \neq 0$. The frame therefore assigns each prime a position that no other prime occupies: prime positions in the frame are unique.

Observation on a non-own circle (`prime_unit_on_other_circle`). A prime p observed on a circle of a different prime modulus q is a unit of that circle’s group: $p^{q-1} \equiv 1 \pmod{q}$, by Lagrange’s theorem applied to the multiplicative group $(\mathbb{Z}/q\mathbb{Z})^\times$ of order $q-1$ (the circle-group size theorem, reused from Exercise XXIV). The prime is a regular point of the foreign circle — it occupies a group element, not the zero — and its position is governed by the circle’s group law.

Observation on its own circle (`fermat_little_complete`). On the circle of its own modulus, a prime satisfies the complete Fermat algebra: for every a , $a^p \equiv a \pmod{p}$. This is the full form of Fermat’s little theorem (coprime case via the

unit group and Lagrange; non-coprime case via congruence lifting), reused from Exercise XXIV. Together with the previous paragraph: on its own circle the prime is the modulus itself (everything is governed by p); on every other circle it is a unit subject to that circle's group.

3 Relation to the exercise series

This exercise connects the two previous ones. Exercise XXIV established the prime circle: a prime p is the modulus of the circle $\mathbb{Z}/p\mathbb{Z}$ and its algebra is Fermat's little theorem. Exercise XXIII established that composites are prime factorizations. The orthogonal frame completes the picture: a composite $n = pq$ is observed as the product circle $\mathbb{Z}/pq\mathbb{Z}$, which the coordinate law factors into the two orthogonal circles $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z}$; the prime factors are the zero-projections of the frame (factor observability), and their circle-algebra is Fermat's theorem. The frame is the geometric form of the Chinese remainder theorem, and the observability theorems are its prime-theoretic reading.

The honest boundary of the exercise is stated as in the previous ones: every theorem is a restatement of classical mathematics (novelty_status: KNOWN), the proofs are machine-checked with zero sorry against mathlib v4.32.2, and no claim is made beyond the statements listed. In particular, the frame changes coordinates; it does not change the distribution of primes.

4 Formalization

The proof is a single Lean file (proof.lean, 272 lines), importing Mathlib.Data.ZMod.Basic, Mathlib.GroupTheory.OrderOfElement, and Mathlib.Data.Nat.Prime.Basic. The seven theorems:

Theorem	Statement	mathlib anchor
orthogonal_frame_crt	$m \perp n \Rightarrow \mathbb{Z}/mn \simeq \mathbb{Z}/m \times \mathbb{Z}/n$	ZMod.chineseRemainder
factor_observable_zero	$p \mid a \Rightarrow (a : \mathbb{Z}/p) = 0$	ZMod.val_injective, mod_eq_zero_of_dvd
distinct_primes_observable	$p \neq q \Rightarrow (p : \mathbb{Z}/q) \neq 0$	Nat.Prime.eq_one_or_self_of_dvd
card_units_zmod_prime	$ (\mathbb{Z}/p)^\times = p - 1$	explicit bijection (Ex. XXIV)
fermat_little	$\gcd(a, p) = 1 \Rightarrow a^{p-1} \equiv 1$	orderOf_dvd_card
fermat_little_complete	$\forall a, a^p \equiv a \pmod{p}$	Nat.ModEq.pow
prime_unit_on_other_circle	$p \neq q \Rightarrow p^{q-1} \equiv 1 \pmod{q}$	Lagrange + CRT-free unit group

Compiled with Lean 4.32.2 / mathlib v4.32.2, zero errors, zero sorry, zero new axioms. The file is self-contained; the circle-group-size and Fermat theorems are restated inside it (reused from Exercise XXIV) so that the exercise does not depend on any other exercise file.

5 Conclusion

The orthogonal Euler frame is a legitimate coordinate system for observing the prime axis: its coordinate law is the

Chinese remainder theorem, its prime observations are Fermat's little theorem and the zero/nonzero projection theorems, and all of it is machine-checked in pure known mathematics. The frame is the geometric form of the CRT, and it connects the composite factorization of Exercise XXIII with the prime circle of Exercise XXIV. Coordinates change; distributions do not — the honest boundary is kept, and the frame's value is observational clarity, not new number theory.

References